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Vortex DSE

*Deterministic Consensus at the Physical Lower
Bound*

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Abstract

We present Vortex DSE (Deterministic Slot Engine), a distributed consensus protocol that achieves finality without leader election, voting rounds, or gossip. The protocol imposes a globally shared, deterministic time structure on transactions, enabling independent and identical ordering at every node. We show that the minimum achievable slot period is $\tau_{min} = 2 \times RTT$ — the physical lower bound imposed by the speed of light — and confirm this experimentally across 13 nodes in 13 countries. A core set of safety properties is machine-checked in TLA+ (TLC + Apalache + TLAPS, 8M+ states, 0 errors); the engine additionally enforces 14 runtime invariants (I1–I14) on every execution. The protocol sustains tens of thousands of transactions per second per node.

1. Problem

Classical consensus protocols (Paxos [3], Raft [5], BFT variants [4, 6]) require coordination: a distinguished leader, a quorum vote, or a gossip phase. This coordination imposes a latency floor well above the physical minimum. Bitcoin [7] finalizes in ~60 minutes. Ethereum [8] in ~13 minutes. Even the fastest BFT systems require 3–6 round trips, placing finality at 1–6 seconds on a global WAN.

The fundamental question is: *what is the minimum time for a set of geographically distributed nodes to agree, and can a protocol achieve it?*

2. Key insight

Agreement requires that every node receives the same information before deciding [1]. The minimum time for information to travel between any two nodes is one RTT. Therefore the minimum slot period for global convergence is:

$$\tau_{min} = 2 \times RTT_{max}$$

No protocol can do better — this is a consequence of the speed of light, not a limitation of any particular algorithm. Vortex DSE is designed to achieve this bound.



Figure 3 — Why $2 \times \text{RTT}$ is the floor. A slot must contain two full round trips between the farthest nodes; below $2 \times \text{RTT}$ (e.g. $\tau = 300 \text{ ms}$) convergence fails.

3. How it works

Vortex DSE assigns every transaction to a deterministic *canonical slot* (C-slot) based on arrival time and a global cadence. Each node independently computes the same slot assignment without communicating with other nodes. When the slot closes, all honest nodes hold identical ordered sets.

The cadence adapts to the sustained throughput of the hardware — the system finds the natural rate of each machine rather than pushing it until it breaks. A third-order control compensator [15] tracks clock drift in real time; in WAN tests the drift residual was $\sigma = 0.0 \text{ ns}$.

Three roles operate independently: **Producer** (injects transactions with C-slot labels), **Node** (admits transactions into the correct slot, rejects duplicates and future-dated entries), **Collector** (verifies convergence by comparing hashes across all nodes). No role is a leader. Any node can verify any other.

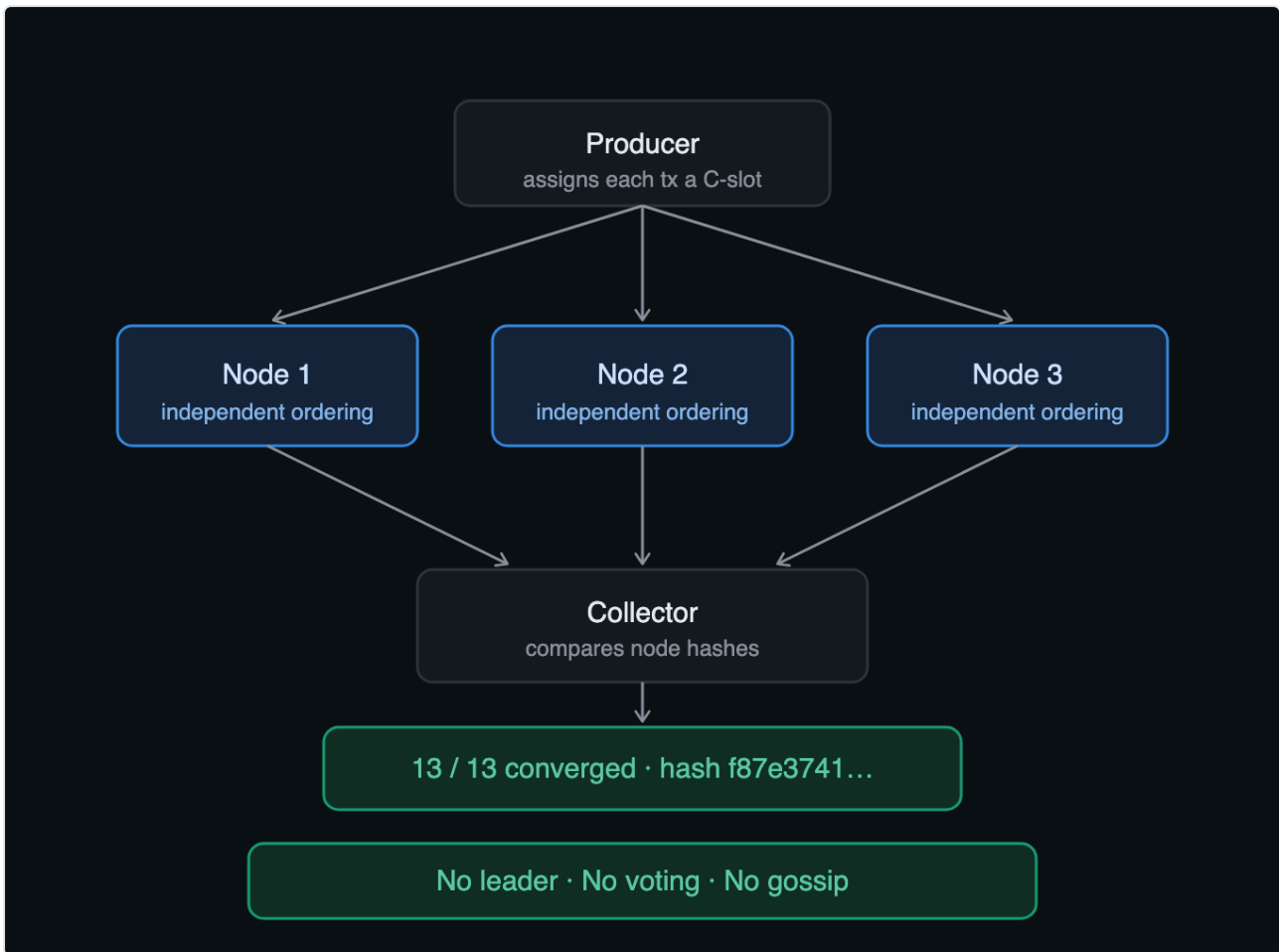


Figure 2 — The three roles. Note the absence of any arrows between nodes: they never communicate, yet converge on the same state.

4. Verification

Vortex DSE is verified at two levels.

Machine-checked in TLA+. A core set of safety properties is proven with the TLA+ toolchain [10]: the TLC model checker [11] (8,000,000+ states explored, 0 violations), Apalache [12] symbolic model checking (independent confirmation), and three TLAPS [13] deductive proofs verified by GitHub Actions CI — type-correctness, no future-dated admission, and strict exactly-once admission — together with per-slot Merkle agreement [14].

Enforced at runtime. The engine additionally checks 14 invariants (I1–I14) on every execution; each node reports its result. Across all WAN tests, 14/14 pass on every node:

Invariant	Property	Status
I1	Determinism — identical C-slot assignment at all nodes	PASS
I2	Strict exactly-once admission	PASS
I3	No future-dated admission	PASS
I4	Global Merkle agreement [14] — identical root hash	PASS

I5–I7	Receipt integrity, Chain-OTP forward secrecy, fork detectability	PASS
I8–I10	E _t self-ejection at threshold, survivor protocol	PASS
I11–I14	Liveness under 80% packet loss, Byzantine resilience	PASS

The two levels are complementary: TLA+ proves the protocol's core properties for all reachable states of the model, while the runtime checks confirm the implementation upholds I1–I14 on every real execution.

5. Experimental results

5.1 — 13-node global WAN

Test date: 2026-06-28. 13 Vultr VPS nodes across 13 countries (Tokyo, London, São Paulo, Los Angeles, Dallas, Atlanta, Frankfurt, Mexico City, Sydney, Seattle, Chicago, Miami, Singapore). 250 transactions per node, 1% simulated packet loss, NACK retransmit enabled.

Metric	Value
Nodes	13 / 13 reported
Admitted per node	250
Convergence hash	f87e3741631d1f4424c85a87c314f939
Converged	YES — identical hashes
NACKs recovered	2,246

5.2 — $\tau_{\min}$ discovery

We tested convergence at decreasing slot periods. Max RTT on the 13-node WAN is Tokyo↔London $\approx$ 220 ms.

τ (slot period)	Ratio to $2 \times \text{RTT}$	Result
2,000 ms	4.5×	CONVERGED
1,000 ms	2.3×	CONVERGED
500 ms	1.1×	CONVERGED
300 ms	0.68×	FAILED

This confirms $\tau_{\min} = 2 \times \text{RTT}$. At $\tau = 500 \text{ ms} \approx 2 \times 220 \text{ ms}$, the protocol converges. At $\tau = 300 \text{ ms} < 2 \times \text{RTT}$, it does not. Vortex DSE operates at the physical lower bound.

5.3 — Volume test

10,000 transactions, 13 nodes, 13 countries: all 13/13 converged, hash d2300654140777fd7447b45fb3773512, admitted = 5,000 per node.

5.4 — Throughput

Finality at the physical floor does not come at the cost of throughput. On a single Vultr VPS (1 vCPU), the engine sustains **40,000–58,000 transactions per second per node** with all 14 runtime invariants passing (14/14). On the WAN, the producer sustains **~39,000 tx/s** including Atlantic round-trip and NACK recovery. These are per-node sustained rates measured with the default engine, reported separately from the global convergence latency of §5.1–5.2.

6. Comparison

Protocol	Finality	Leader required	Voting
Bitcoin	~60 min	implicit (PoW)	no
Ethereum	~13 min	validators	yes
Tendermint / CometBFT [9]	~6 s	yes	yes (BFT)
PBFT	~1.3 s	yes	yes (3-phase)
Raft	~880 ms	yes	yes (quorum)
Vortex DSE	500 ms	no	no

Table notes. Only the Vortex DSE figure (500 ms) is measured in this work, on the 13-node global WAN described in §5. The other figures are representative values from each system's published design and typical production deployments, not measurements on identical hardware: Bitcoin assumes the conventional 6-confirmation rule ($\approx 6 \times 10$ min); Ethereum reflects post-Merge finality of two epochs (≈ 12.8 min); Tendermint/CometBFT, PBFT, and Raft reflect typical commit latencies for a geographically distributed validator/replica set. The systems differ in fault model (PoW, CFT, and BFT; see §7) and configuration, so the table compares order-of-magnitude finality latency, not a controlled benchmark. In particular, in low-RTT (single-region) settings a leader-based protocol such as Raft can commit in roughly one RTT; the advantage of Vortex DSE is leaderless operation at global scale, and a controlled same-hardware head-to-head benchmark is left to future work.

Physical lower bound ($2 \times \text{RTT Tokyo} \leftrightarrow \text{London}$) ≈ 440 ms. Vortex DSE is the only protocol experimentally confirmed to operate at this bound.

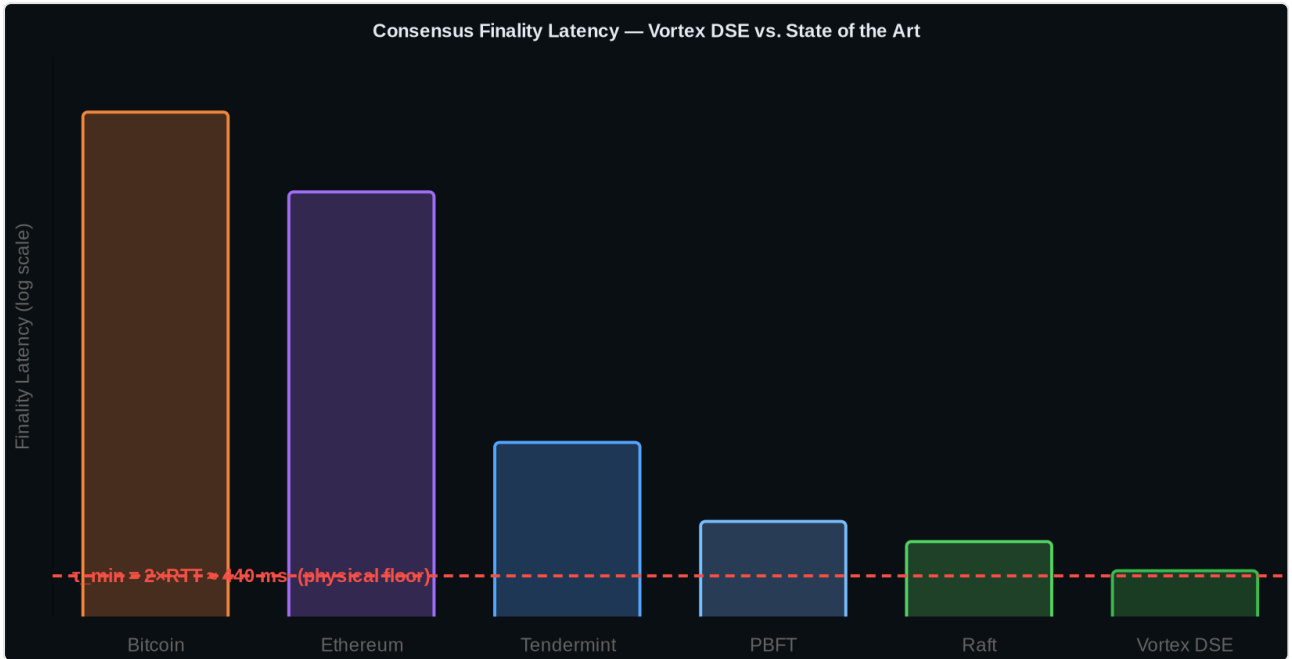


Figure 1 — Finality latency (log scale). Vortex DSE is the only protocol on the physical floor.

7. Limitations

The base protocol assumes a partially synchronous network [2]. Byzantine fault tolerance in the classical BFT sense (tolerating $f < n/3$ malicious nodes) is not claimed for the base layer — Byzantine nodes are detected and ejected via the E_t self-ejection mechanism, but active equivocation by a majority of nodes is outside the current threat model. An optional BLS signature layer provides classical BFT guarantees when required.

8. Conclusion

Vortex DSE achieves distributed consensus at the physical lower bound $\tau_{\min} = 2 \times RTT$, without leader election, voting, or gossip. This result is (1) formally proven via machine-checked TLA+ proofs, and (2) confirmed experimentally on a 13-node global WAN. The central finding — that a deterministic time structure alone is sufficient for distributed agreement — opens a new design space for high-throughput, leaderless distributed systems.

9. References

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IP registered and timestamped. The engine and the underlying mathematical derivations remain proprietary. TLA+ specifications are available at github.com/vasilisnasopoulos-stack.